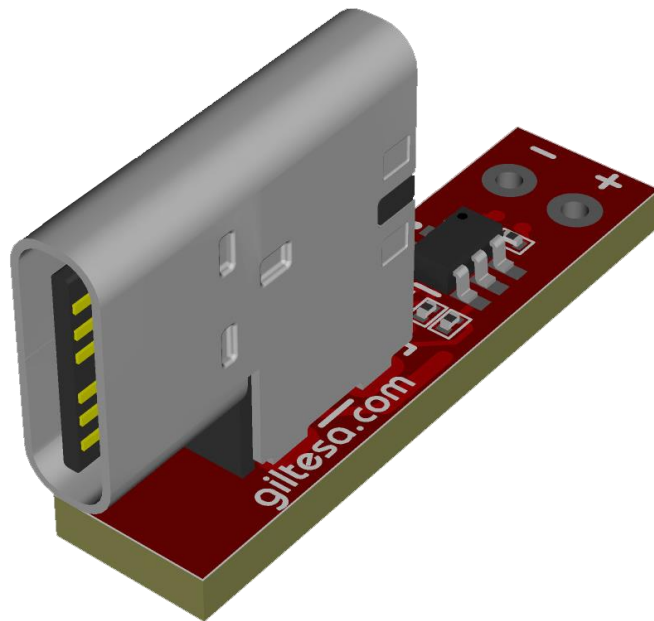


USB-C KIT FOR SUPER NINTENDO



PRODUCT

[HTTPS://SHOP.GILTESA.COM/PRODUCT/SUPER-NINTENDO-USB-C-KIT](https://shop.giltesa.com/product/super-nintendo-usb-c-kit)

PLEASE READ THROUGH THESE INSTRUCTIONS ENTIRELY BEFORE ATTEMPTING TO INSTALL

WARNING: IF YOU ARE NOT COMFORTABLE WITH SOLDERING, OR PERFORMING ANY STEP IN THIS GUIDE, DO NOT PERFORM THE INSTALL YOURSELF. FIND SOMEONE WHO IS COMFORTABLE TO DO IT FOR YOU.

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DESCRIPTION

The **Super Nintendo: USB-C Kit** is a board that allows to replace the original power connector for a modern and standard USB-C.

If your original connector is too old or damaged and you need a new one, or if you would like to power up your Super Nintendo with a standard USB-C charger, such as the charger for your Nintendo Switch, phone, or laptop, you can do so with this kit.

FEATURES

- Exact shape for Super Nintendo / Super Famicom.
- External power through USB-C. ⁽¹⁾
- 5V or 9V power output to supply the SNES (from version 2.0).

⁽¹⁾ The USB-C requires trimming of the shell.

INCLUDED

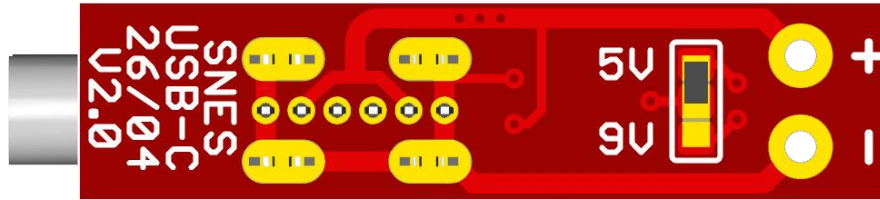
- 1 USB-C board.
- 1 Power cable.
- 1 Plastic cap for the USB-C board.

RECOMMENDED / REQUIRED [NOT INCLUDED]

- 4.5mm gamebit and phillips screwdrivers.
- Cutting plier.
- Cutter.
- Tin soldering iron.
- Tin.
- Flux.
- Isopropyl alcohol.

BOARD DETAILS

There are 2 pads to solder the included cable. The following explains what each pad is for.



- **VCC:** The +5V line output to power the SNES main board.
- **GND:** The ground pad.
- **Solder Jumper:** 5V or 9V power output selector to supply the SNES. ⁽²⁾

HOW DOES IT WORK?

The USB-C Power Delivery technology (USB-C PD) allows for communication between the device and the power supply(charger), enabling the negotiation of the voltage to be supplied through the USB-C cable. In this case, the microchip included on this board is configured to request 5V or 9V⁽²⁾, signaling to the charger that it requires 5V or 9V⁽²⁾ to run on. If the charger is compatible, it will supply the requested voltage. If it's not compatible, nothing will happen, and the console won't power on.

⁽²⁾ Version 2.0+ of this board has a selection solder jumper; if you solder it in one position, the circuit will require 5V, and if you solder it in the other position, it will require 9V from the power supply.

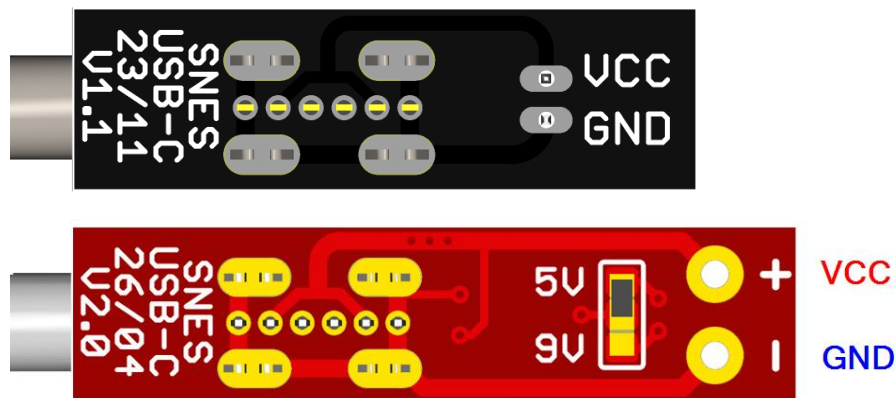
By default, it is configured to supply 5V.

TEST THE BOARD!

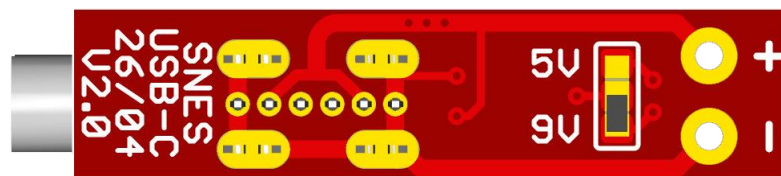
Before starting the installation, you should test the board. If it doesn't work contact me [for a replacement](#) (all boards are fully tested, but they may damage during the shipping, we try to package them as better as possible), if it works, go ahead with the installation.

There are two board versions: version 1.X (older), which always works at 5V, and version 2.X (newer), which by default works at 5V but can be configured to 9V.

Connect the power from your USB charger to the USB-C connector on the board. Then, using a multimeter in voltage measurement mode, **check for a 5V** reading at the **VCC** and **GND** pads.



If your board is the new version and you are going to use 9V power, you can change the solder jumper now from 5V to 9V by moving the solder bridge and then test with a multimeter to confirm that you get 9V.



INSTALLATION STEPS

Please, carefully read the following steps for a successful installation.

PRE INSTALLATION STEPS

Before the installation, your SNES may need some extra steps to have it ready for the kit.

1. DISASSEMBLY THE SUPER NINTENDO

Nintendo products in general use two kind of screws. The first one called **gamebit** to close the shell, and the second one called **phillips** to hold the main board to the shell.

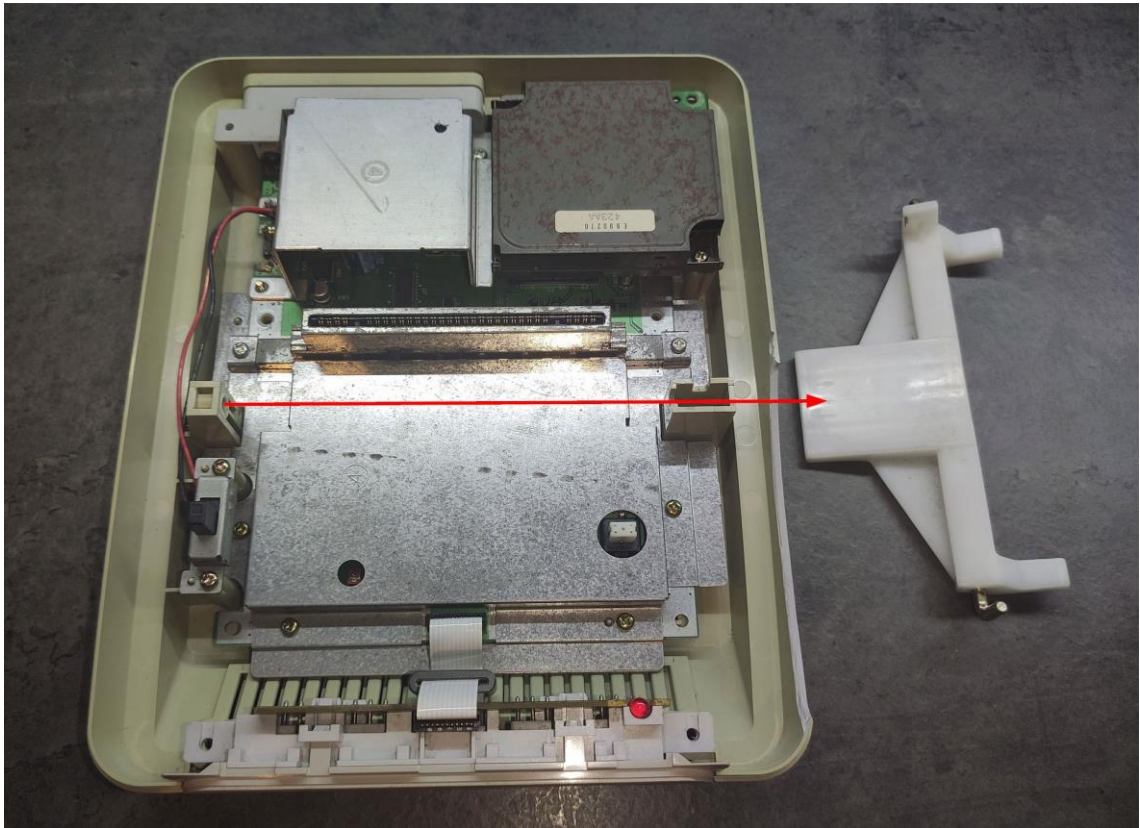


Gamebit screwdriver

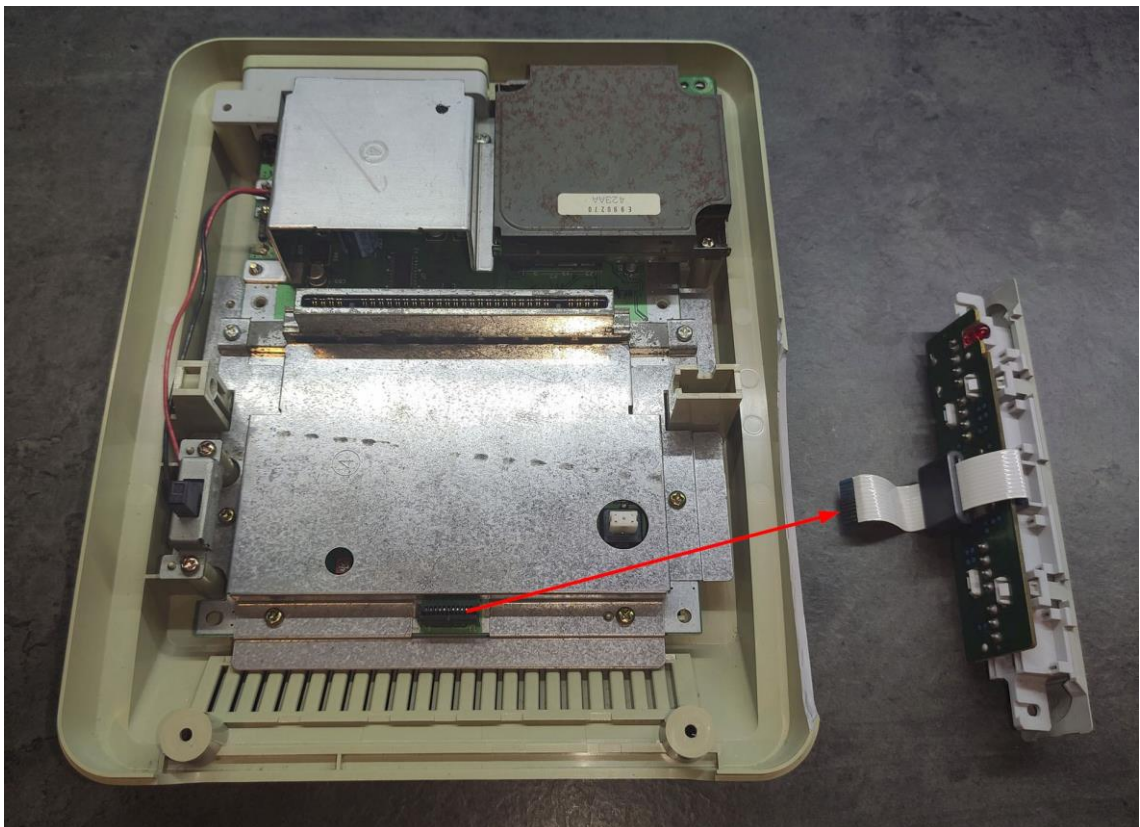
Use the 4.5mm **gamebit** screwdriver to open the shell and remove the 6 **gamebit** screws.



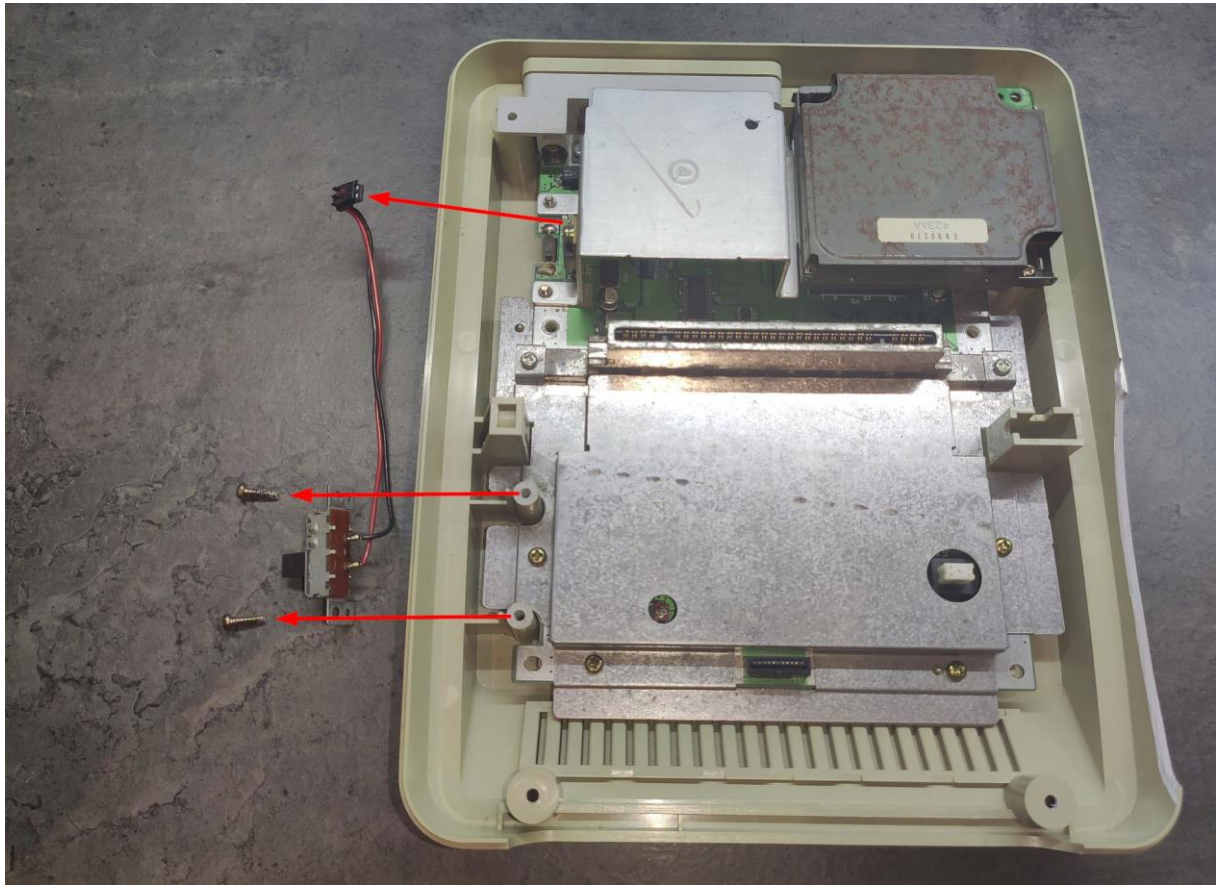
Remove the white plastic piece that allows the game cartridges to be ejected.
Be careful not to lose the spring.



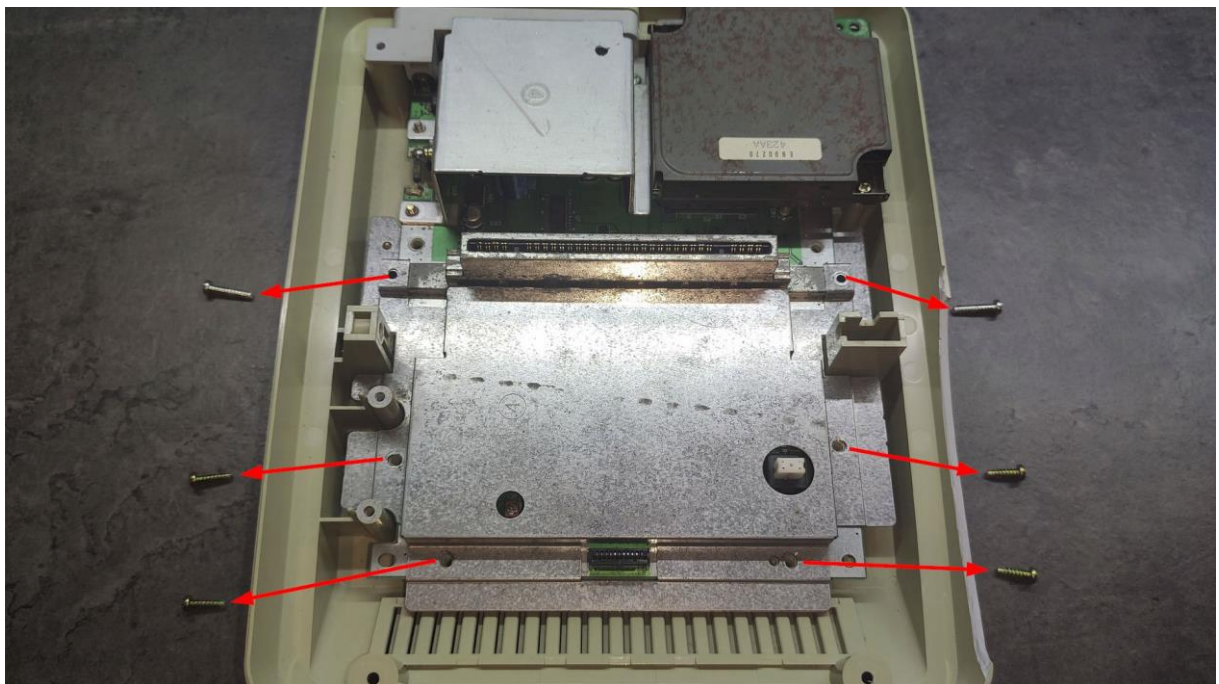
Disconnect the ribbon cable from the front panel where the game controls are connected.



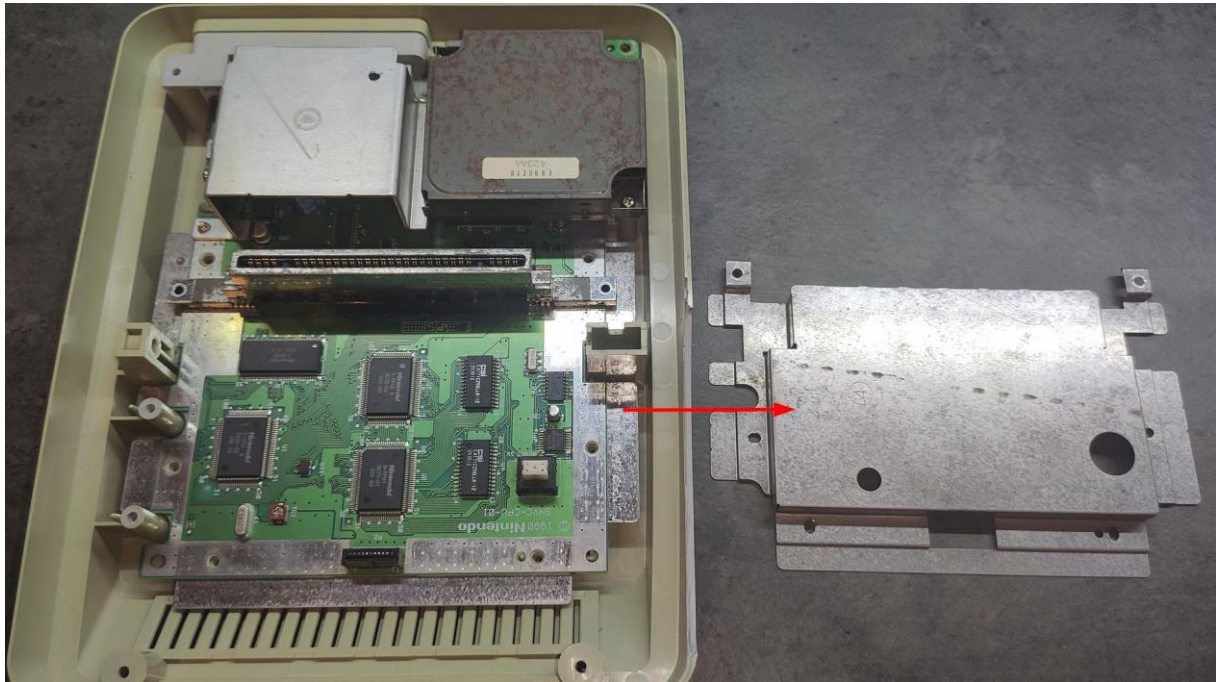
Remove the two screws from the power switch and disconnect it from the mainboard.



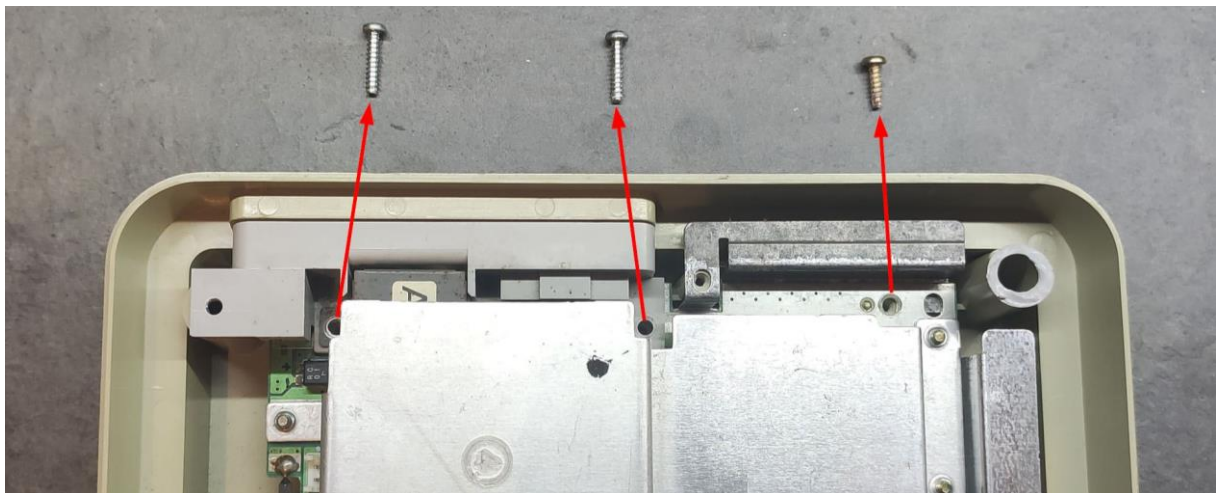
Then remove the six screws holding the top metal plate.



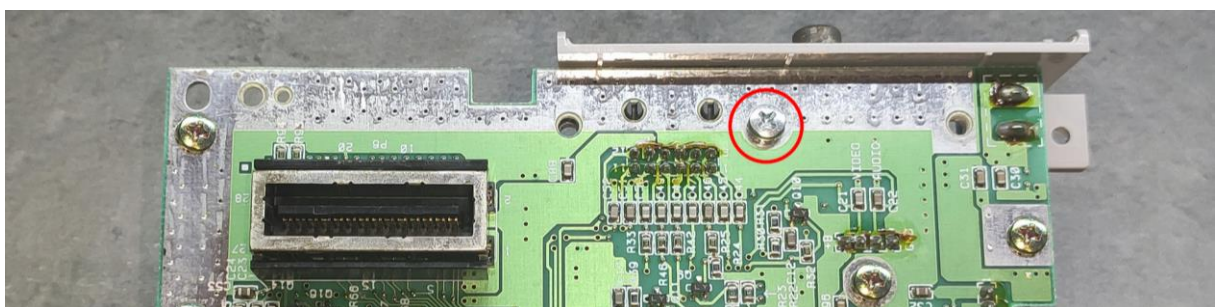
Remove the metal plate as well.



Remove these three screws that hold the motherboard to the lower plastic shell.



And this other screw on the underside of the motherboard.

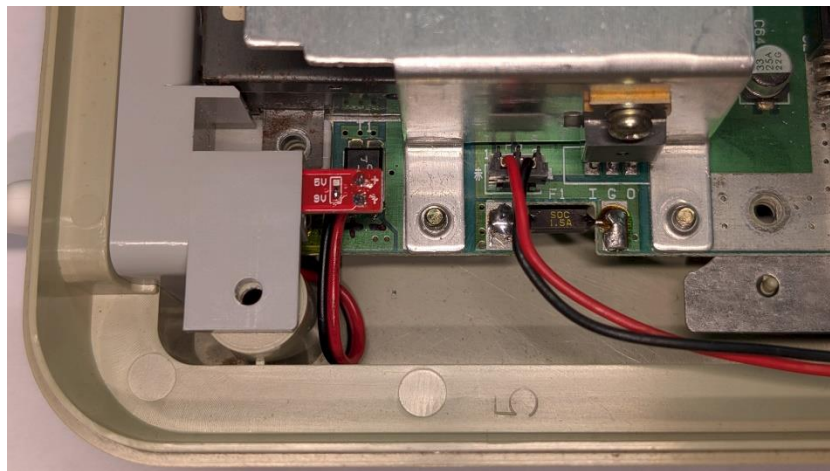


2. REMOVE UNNECESSARY PARTS

The installation can be done in two different ways depending on the version of the USB-C board and how you want to power the console.

Power supply at 9V, like the original power supply

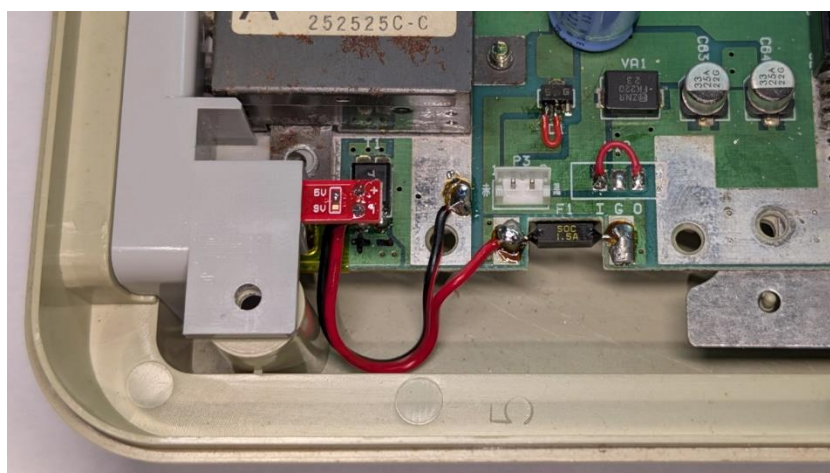
For the red board, version 2.X, it is possible to power the console at 9V, just like Nintendo's original power supply does. In that case, you only need to replace the DC jack and nothing else.



9V power supply installation complete

Power supply at 5V

The console operates internally at 5V, so you can remove the voltage conversion component and power it directly at 5V from USB-C.



5V power supply installation complete

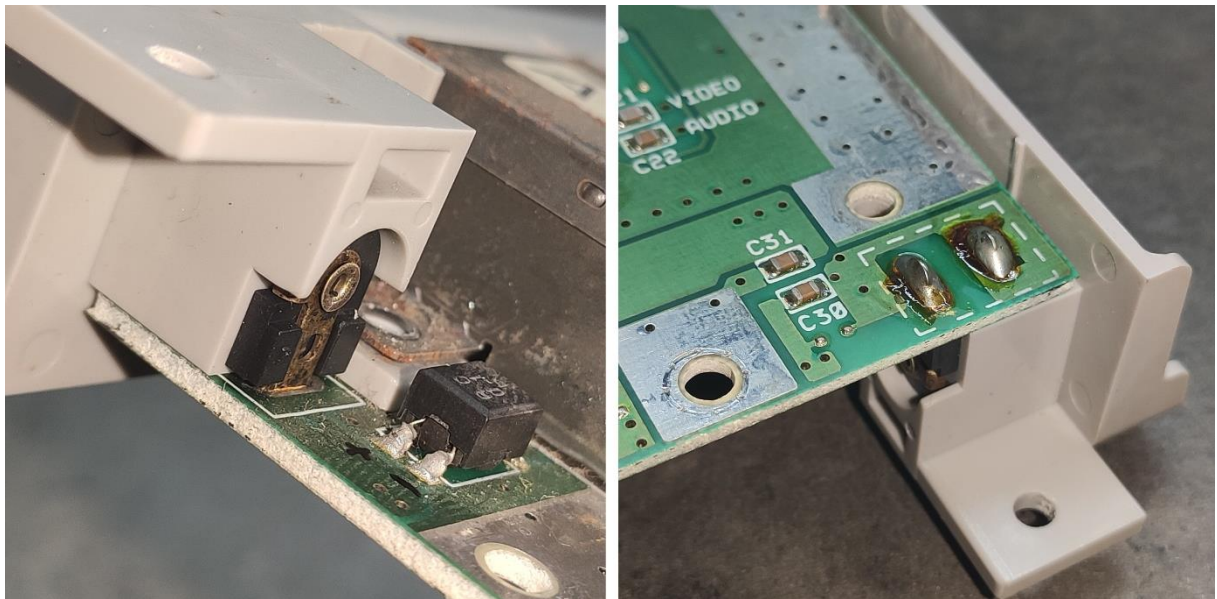
Next, the installation is explained step by step.

DC POWER JACK CONNECTOR

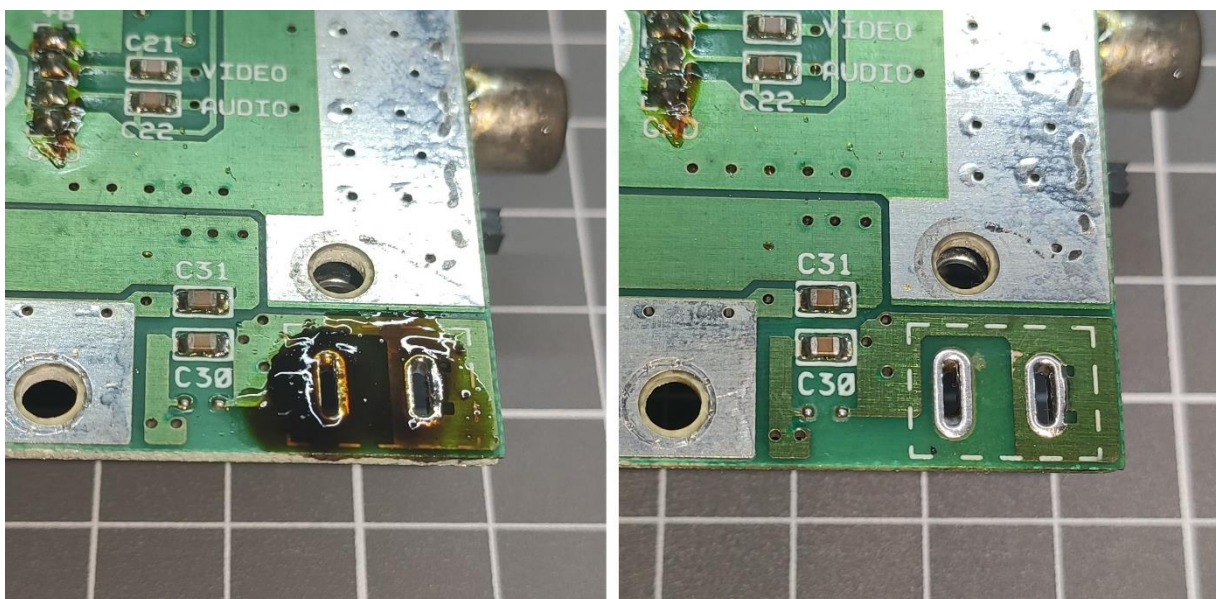
Both installation methods require removing the DC power jack connector.

The connector is embedded in the rear plastic, meaning both are a single piece and cannot be separated without breaking the connector.

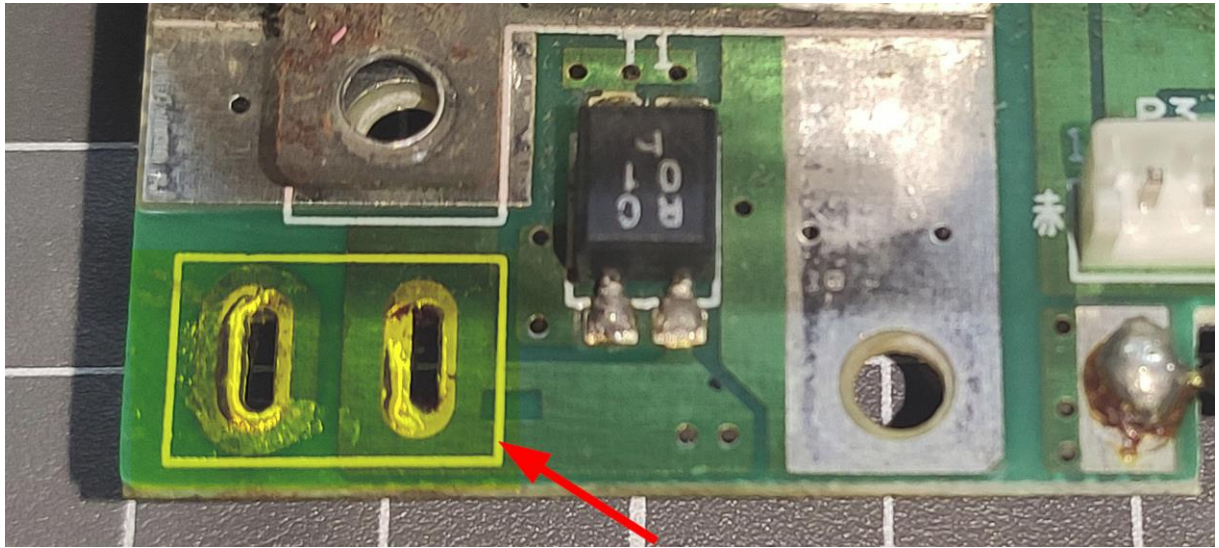
To remove the connector, heat both lower pads so the solder melts, then remove the connector and the plastic at the same time. The connector will likely break, and part of the plastic may also melt. This is not a problem, as further modifications will be required in later steps.



After removing the connector, clean all flux residue with alcohol.



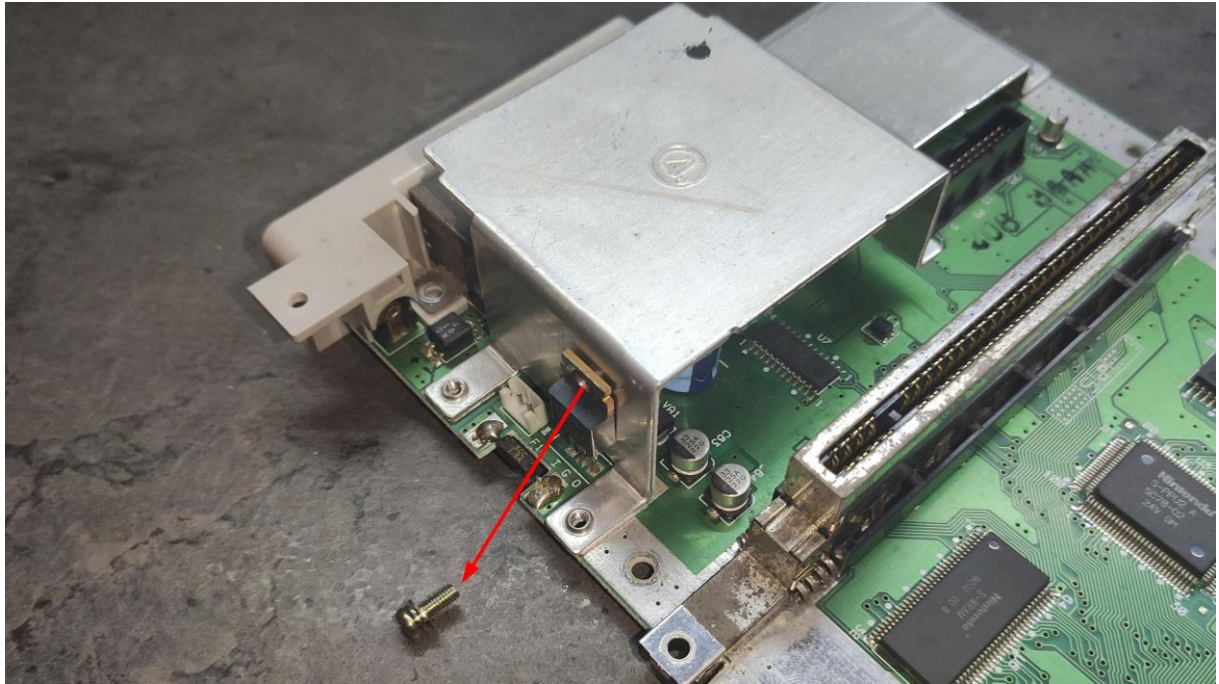
Apply Kapton tape in this area, as the USB-C connector could touch both pads and cause a short circuit.



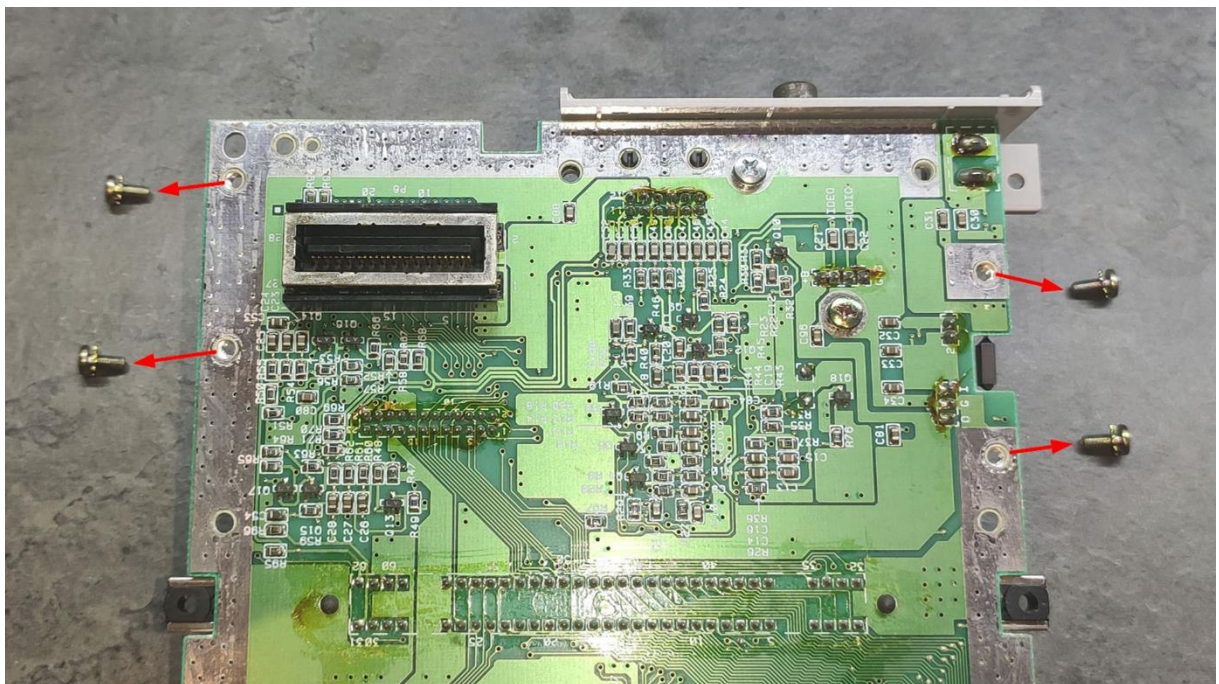
VOLTAGE REGULATOR AND OTHER COMPONENTS

This is only necessary when powering the console at 5V (black board always, or red board optionally).

Remove the side screw that secures the DC-DC converter. You can also remove the aluminum heatsink, as it will no longer be needed, by removing the four screws.

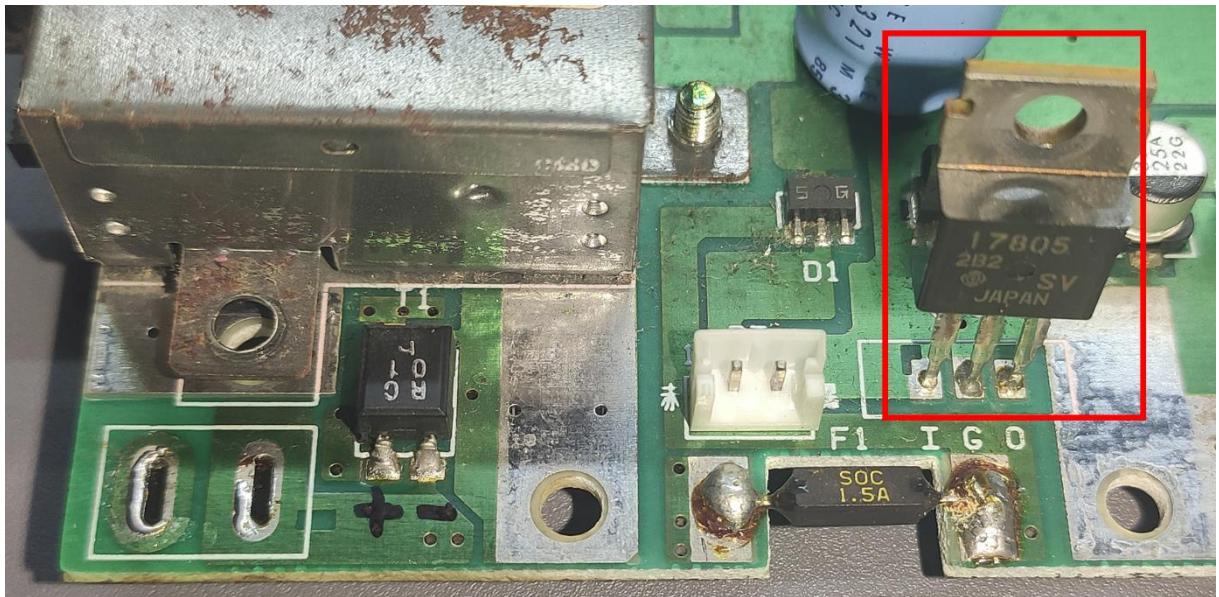


Only for 5V power supply.

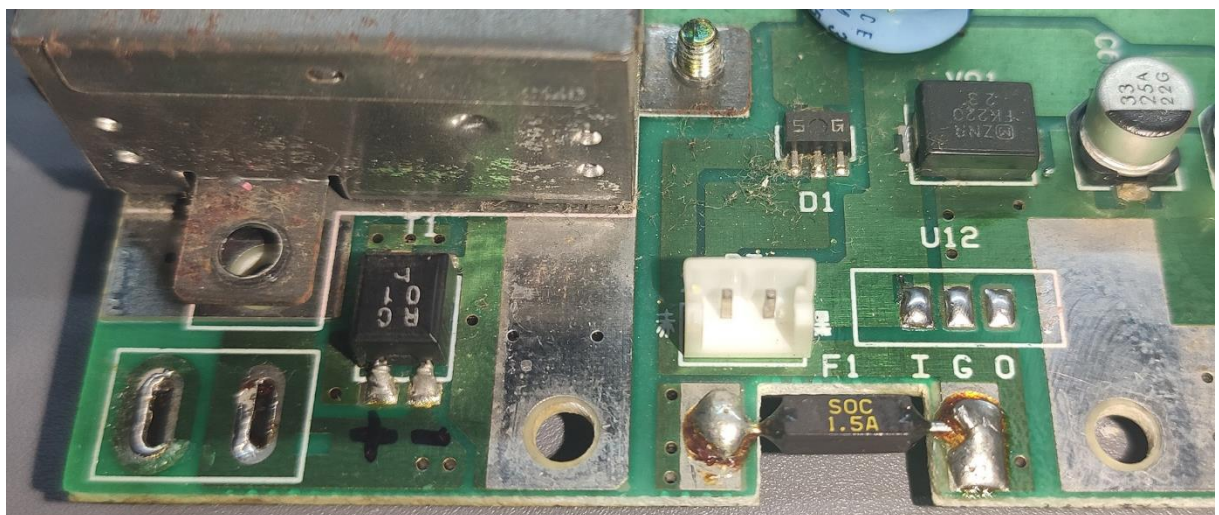


Only for 5V power supply.

Now it is necessary to bridge the input and output pins on the pads where the converter is soldered. For a cleaner installation, the component can be removed first, and then the bridge can be applied more easily and neatly.

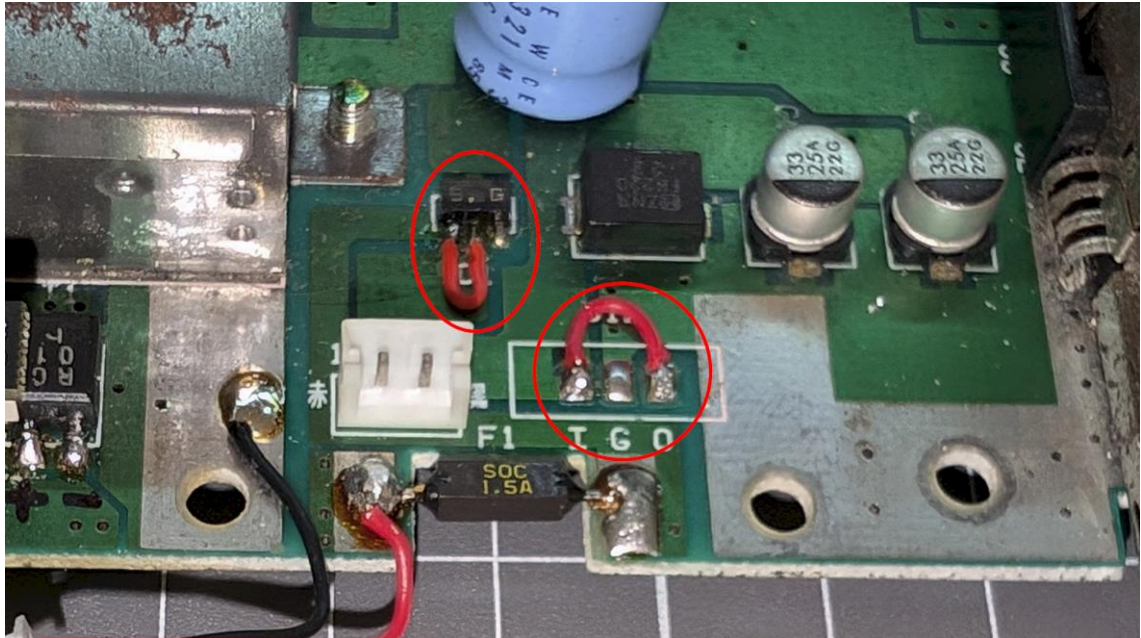


Only for 5V power supply.



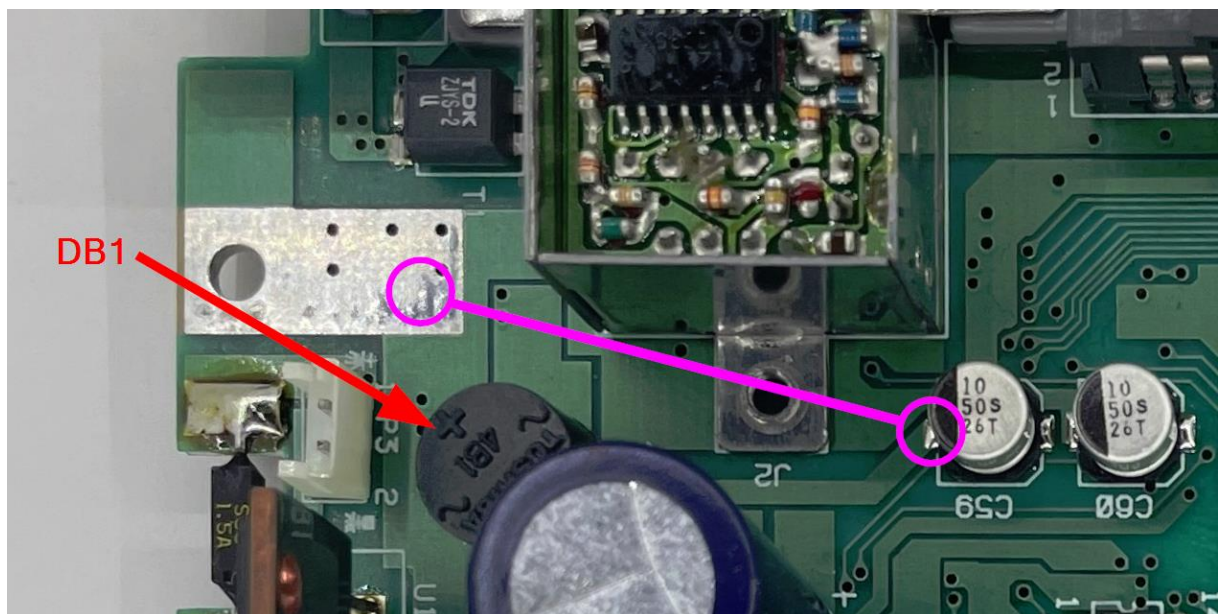
Only for 5V power supply.

After removing the component, in our case, the mentioned bridge must be applied. It is also necessary to apply another one on diode D1. For this last one, only the bridge is applied without removing the component, since it is small and difficult to remove.



Only for 5V power supply.

If your mainboard is the SNSP-CPU-01 or SNSP-CPU-02 model, you can confirm it by checking for the presence of the **DB1** component. In this case, you'll need to add one more wire, connecting the two points marked in pink.



Only for 5V power supply and SNSP-CPU-01 or SNSP-CPU-02 mainboards.

INSTALLATION STEPS

1. REMOVE THE CONNECTOR

When desoldering the connector from the mainboard, the plastic may have melted. This is not a problem because we will need to trim certain parts to make space for the new connector.

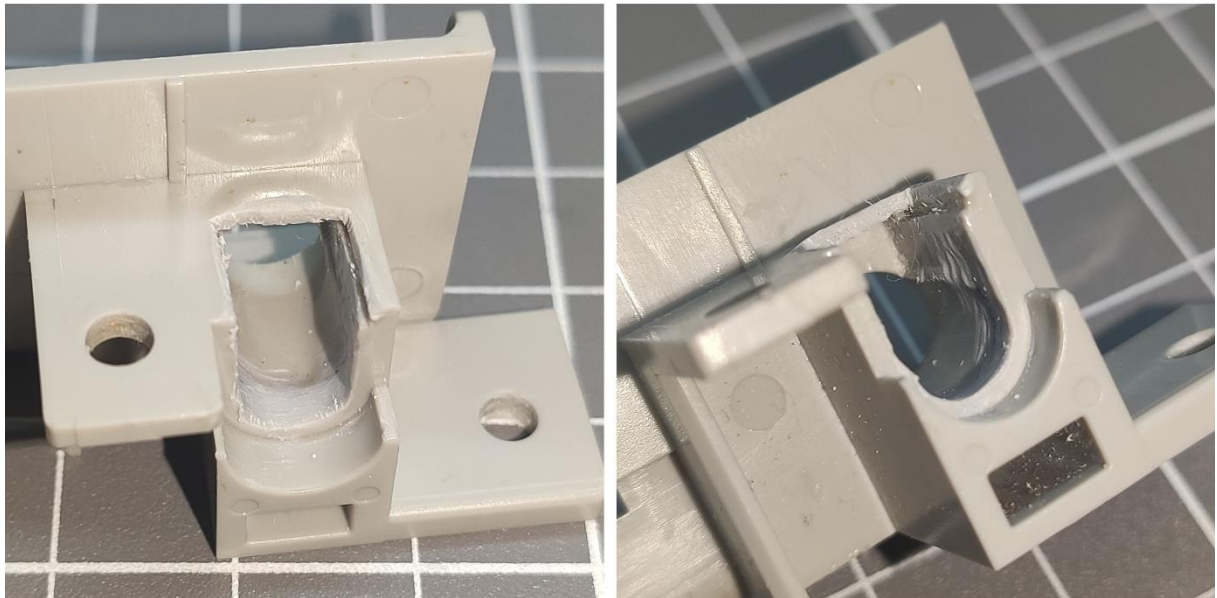
Remove the connector parts using cutting pliers.



2. MODIFY THE PLASTIC

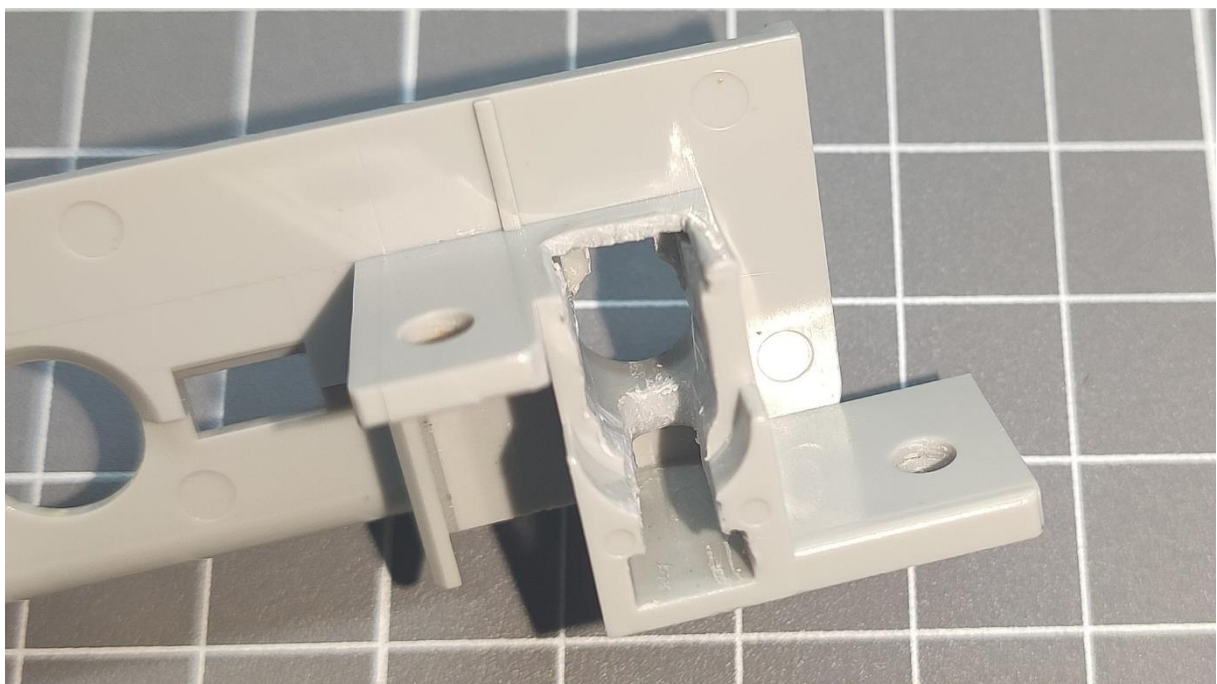
Trim the plastic to make space for the USB-C board. First, remove the plastic from the sides so it looks like in these images.

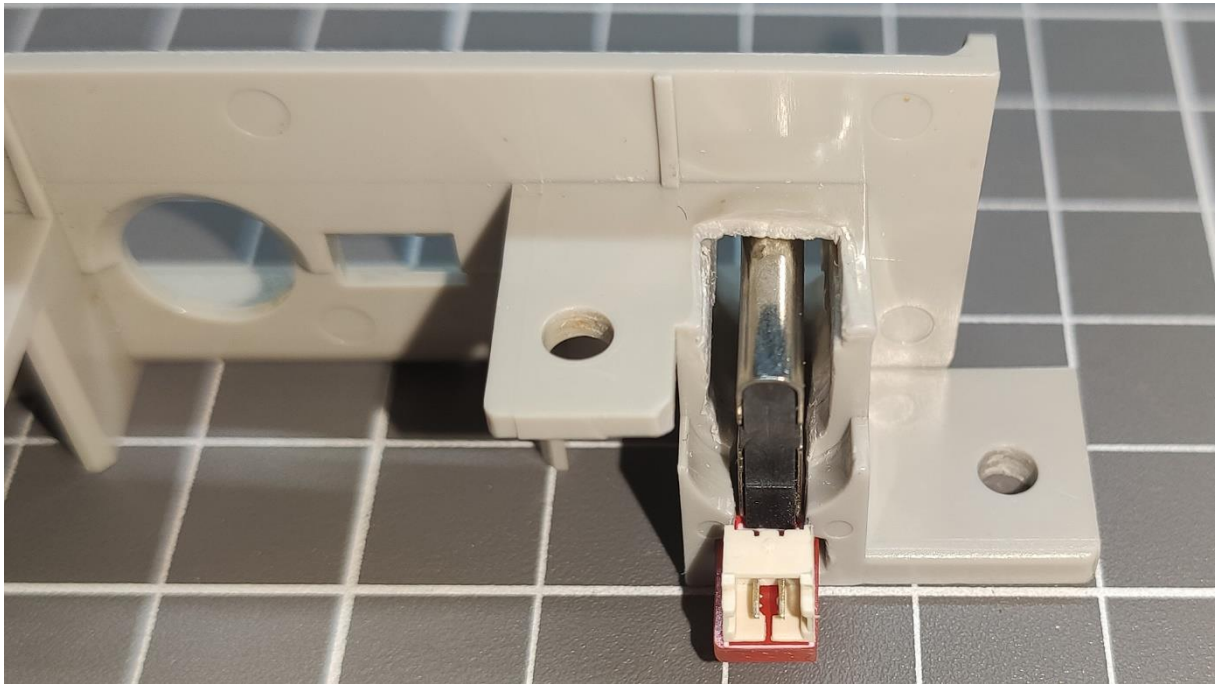
Don't worry if it's not perfect, this part cannot be seen from the outside.



Afterward, a groove needs to be made connecting the main hole of the connector with the rectangular hole located just above it. This groove should be made in the center and should be about 3.3 mm wide.

This will allow the circuit board to be placed inside the rectangular hole and the USB-C connector to sit inside the main connector opening.





Finally, enlarge the front hole so that the USB-C can protrude. Place the board in position, and you will notice that the hole is not large enough. Gradually file the plastic and test with the board until it is large enough.



Don't forget to place the plastic cap to cover the hole and achieve a perfect finish:



If you make the plastic cuts very tight, the connector will be firmly secured. Otherwise, you can apply a small amount of glue to fix it to the plastic.

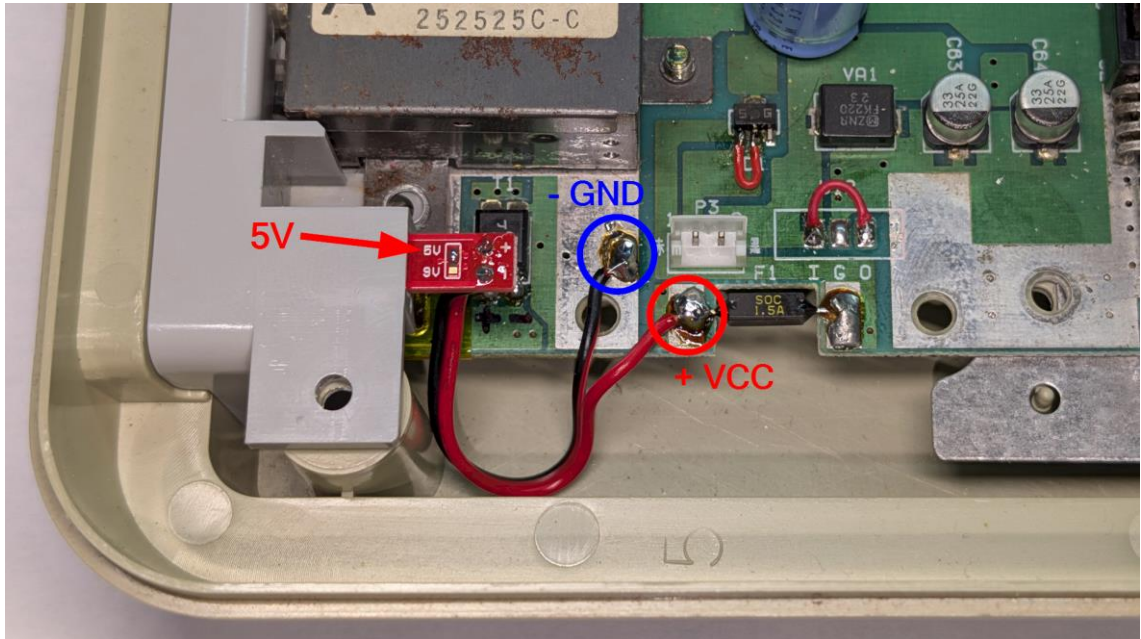


3. ELECTRICAL INSTALLATION

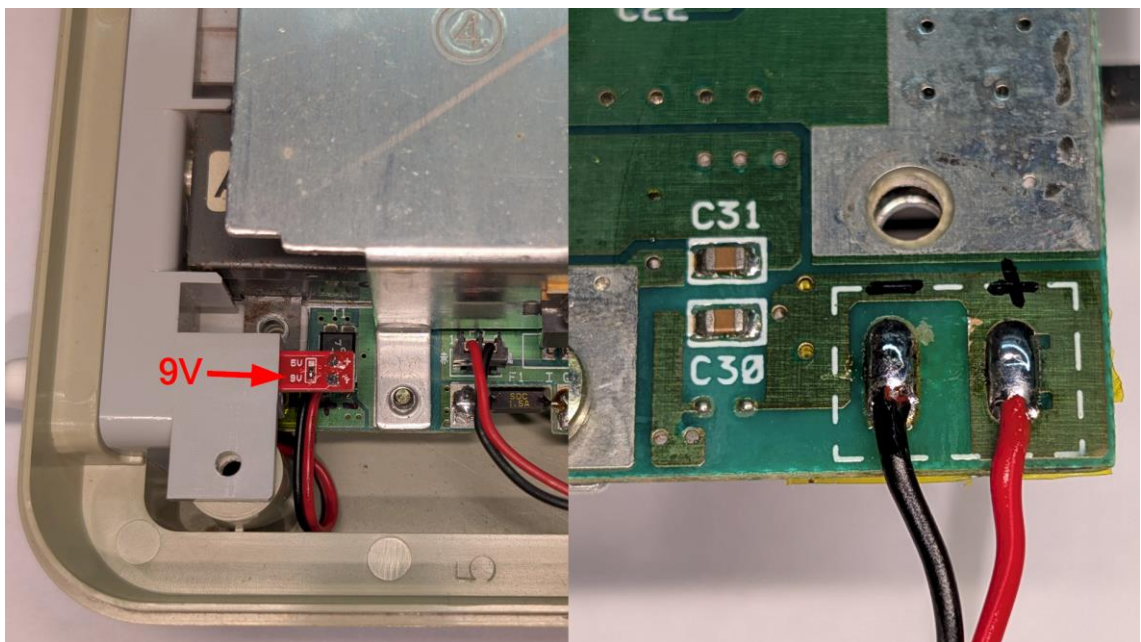
The electrical installation is practically complete; only the included cable needs to be soldered between the USB-C board and the mainboard.

Depending on whether the installation is 5V or 9V, the cable is soldered in a different location.

Power supply at 5V



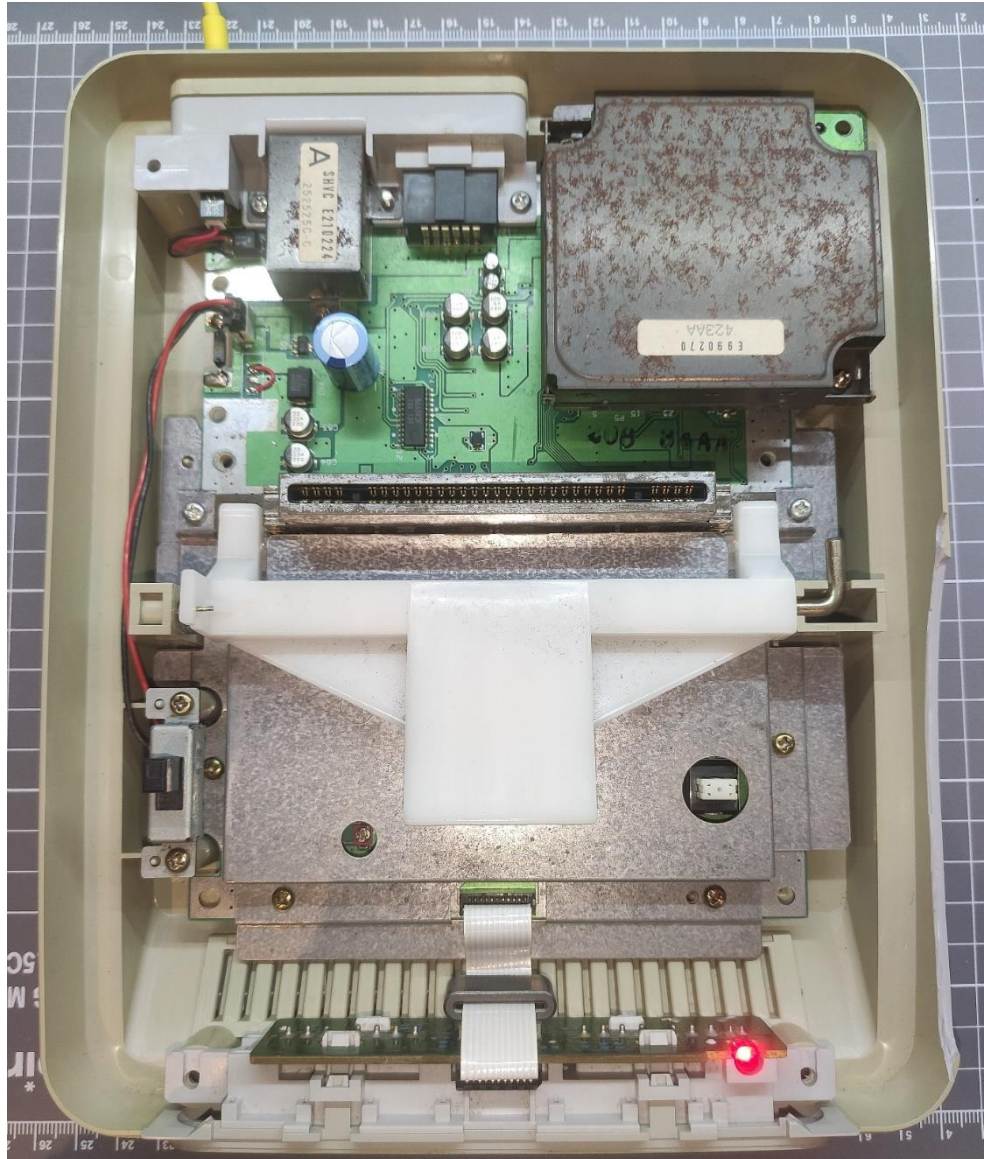
Power supply at 9V



The cable is soldered on the underside of the mainboard. Make sure the polarity is correct for your console. In my case, the GND pin is the one positioned furthest inward.

4. **DONE!**

Follow the steps backward, placing each thing in its place, and the installation will be complete!



FREQUENTLY ASKED QUESTIONS - FAQ

WHAT CHARGER CAN BE USED?

For the 5V installation, you can use any standard charger for mobile phones, computers, etc., with 5V 1A. It does not need to be a Power Delivery charger since this feature is not used.

For the 9V installation, you need a Power Delivery charger.

Technical data for curious minds:

Power Delivery chargers can supply a wide range of voltages: 5V, 9V, 12V, 15V, and 20V. However, for this to happen, the device must communicate with the charger to explicitly request the desired voltage. Without this communication, the charger will never supply more than 5V. That's one of the advantages of USB-C, as it can be used with both old and modern devices.

IS THERE A RECOMMENDED POWER SUPPLY?

Any USB-C Power Delivery charger with a **9V 1A** output is more than sufficient. We used the [Ikea SJÖSS 45W](#) (9V 3A) for all our tests.



MY POWER SUPPLY DOESN'T HAVE A 9V OUTPUT.

Power Delivery chargers can supply a wide range of voltages: 5V, 9V, 12V, 15V, and 20V. However, they're not required to support all of them, a charger might offer 5V, 9V, and 15V but not 12V or 20V.

That's why it's important to check whether your charger actually supports 9V output.

CAN THIS KIT BE USED WITH OTHER MODS?

Of course, as long as you use the appropriate power supply, you can use any other mod or accessory you would normally use with the original power supply.

However, if you use a special mod that requires direct 9V power, and your installation is set to 5V, you will not be able to use it with this kit.